

PCCST501 - COMPUTER NETWORKS

Assignment 3

Module 3 - Data-Link Layer, Wireless LANs & Mobile IP

LAN, Ethernet & Wireless Network Investigation (CO3)

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Contents

- Q1. Data-link layer responsibilities; LLC vs MAC
- Q2. Ethernet frame format
- Q3. Wireshark capture of Ethernet frames
- Q4. Switch operation and MAC learning
- Q5. Hub vs switch vs bridge
- Q6. CSMA/CD and its reduced role today
- Q7. CSMA/CA in IEEE 802.11 (compared with CSMA/CD)
- Q8. Wireless LAN design for a college laboratory
- Q9. Mobile IP
- Q10. Wired Ethernet vs wireless LAN frame comparison
- References

Q1. Responsibilities of the data-link layer; LLC vs MAC

The data-link layer moves a datagram across **one link** - from a node to the next adjacent node (host to switch, router to router, station to access point). The network layer handles the end-to-end path; the data-link layer handles each hop. TCP/IP defines no link protocols of its own, so it uses whatever the attached network offers (Ethernet, Wi-Fi, PPP and so on).

Main responsibilities

- **Framing** - encapsulating the network-layer packet with a header and trailer so that the receiver can find where a frame starts and ends.
- **Link-layer (physical) addressing** - source and destination MAC addresses identify nodes on the same link; ARP maps an IP address to a MAC address.
- **Error detection (and sometimes correction)** - a checksum or CRC in the trailer lets the receiver detect corrupted frames; Ethernet simply discards them, 802.11 recovers with ACK and retransmission.
- **Flow control and error control** - used on some links (HDLC, PPP, 802.11 stop-and-wait ACK) so a fast sender does not overrun a slow receiver.
- **Media access control** - deciding who may transmit when the medium is shared (CSMA/CD, CSMA/CA, token passing).
- **Node-to-node delivery over the physical layer** - handing the frame to the physical layer as a bit stream.

LLC versus MAC

IEEE splits the data-link layer into two sublayers so that the same upper-layer interface can sit on top of many different media.

Aspect	Logical Link Control (LLC) - IEEE 802.2	Media Access Control (MAC) - 802.3, 802.11 etc.
Position	Upper sublayer, faces the network layer	Lower sublayer, faces the physical layer
Purpose	Media-independent interface; identifies which network protocol is carried (via SAP/SNAP); optional flow and error control	Frame format, MAC addressing, and the rules for accessing the shared medium
Depends on medium?	No - common to all 802 LANs	Yes - different for Ethernet, Wi-Fi, Bluetooth
Typical functions	Multiplexing network protocols, optional acknowledged service	Addressing, CSMA/CD or CSMA/CA, collision handling, FCS
In practice	Ethernet II uses the Type field instead of an LLC header	Implemented in the NIC hardware/firmware

Q2. Ethernet frame format

Ethernet (IEEE 802.3 / Ethernet II) uses the frame structure below. The preamble and SFD are generated by the physical signalling and are normally not shown by capture tools.

Ethernet II (DIX) frame - field sizes in bytes

Preamble	SFD	Dest MAC	Src MAC	Type/Len	Data + Pad	FCS
7 B	1 B	6 B	6 B	2 B	46 - 1500 B	4 B (CRC-32)

added by NIC hardware

Frame length counted by the standard: 64 B (minimum) to 1518 B (maximum), preamble/SFD excluded

Dest MAC: who should accept it | Src MAC: who sent it | Type/Len: 0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6
Values 1536 (0x0600) and above in this field mean Type; values up to 1500 mean Length (IEEE 802.3).

Figure 1: Ethernet frame format

Field	Size	Purpose
Preamble	7 B	Alternating 1/0 pattern (10101010) that lets the receiver synchronise its clock.
SFD	1 B	Start Frame Delimiter 10101011 - marks the start of the frame proper.
Destination MAC	6 B	Link-layer address of the intended receiver (unicast, multicast or broadcast FF:FF:FF:FF:FF:FF). The NIC compares this with its own address and drops frames not meant for it (unless promiscuous).
Source MAC	6 B	Address of the sending NIC; used by switches to learn which port the sender is on, and by the receiver to reply.
Type / Length	2 B	In Ethernet II it is the Type of the payload (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6). In IEEE 802.3 it is the Length of the data; values of 1536 and above are interpreted as Type.
Data (payload)	46-1500 B	The network-layer packet. Padding is added when it is shorter than 46 bytes so that the frame reaches the 64-byte minimum needed for collision detection.
FCS	4 B	Frame Check Sequence: CRC-32 computed over the address, type and data fields. A mismatch makes the receiver silently discard the frame.

Q3. Wireshark capture of Ethernet traffic

Procedure (to be carried out on the lab PC / personal laptop connected by Ethernet):

- Install Wireshark and select the wired interface (e.g. eth0, enp3s0 or Ethernet) on the capture start screen.
- Start the capture, then generate traffic: ping 8.8.8.8, ping <gateway-IP>, open a web page, and run arp -a.
- Stop after about 30 seconds. Apply the display filter eth (or arp || icmp) to keep readable frames.
- Click a frame, expand **Ethernet II** in the packet-details pane and read *Destination*, *Source* and *Type*. Add columns eth.src and eth.dst via Edit - Preferences - Columns for a single-table view.
- Take a screenshot showing the packet list with at least five frames and one frame expanded.

Sample analysis table (format of the result; **replace the MAC values with those from your own capture**):

No.	Protocol	Source MAC	Destination MAC	Type	Observation
1	ARP	aa:bb:cc:11:22:33	ff:ff:ff:ff:ff:ff	0x0806	Who-has request, destination is broadcast
2	ARP	dd:ee:ff:44:55:66	aa:bb:cc:11:22:33	0x0806	Reply, unicast back to requester
3	ICMP	aa:bb:cc:11:22:33	dd:ee:ff:44:55:66	0x0800	Echo request sent to gateway MAC
4	ICMP	dd:ee:ff:44:55:66	aa:bb:cc:11:22:33	0x0800	Echo reply from gateway

No.	Protocol	Source MAC	Destination MAC	Type	Observation
5	DNS	aa:bb:cc:11:22:33	dd:ee:ff:44:55:66	0x0800	Off-subnet query goes to the gateway MAC, not the DNS server's

Key observation: for traffic leaving the local network, the destination MAC is that of the default gateway while the destination IP is the remote host - MAC addresses change at every hop, IP addresses do not.

Figure 2: Wireshark screenshot - insert here

Packet list with at least 5 frames and the Ethernet II header expanded (store the image as screenshots/q3_ethernet_capture.png in the repository).

Background: Wireshark and tcpdump obtain frames from the datalink layer through libpcap, which hides OS-specific mechanisms such as BPF (BSD), DLPI (SVR4) and PF_PACKET (Linux) (Stevens, *UNIX Network Programming*, ch. 29). The capture library reports the link type (DLT_EN10MB for Ethernet) so that the 14-byte Ethernet header can be skipped to reach the IP header. On a switched network a normal NIC sees only unicast traffic for itself plus broadcast and multicast, unless port mirroring is configured.

Q4. Switch operation and MAC address learning

A link-layer switch forwards frames using the destination MAC address, and builds its forwarding (MAC address) table **automatically** by examining the **source** address of every frame it receives.

Algorithm

- **Learn:** on receiving a frame on port p with source address S , record (S, p) in the table, or refresh its ageing timer (typically 300 s).
- **Look up** the destination address D . If D is found on port q and q is not the incoming port, **forward** only to q .
- If D is found on the **same** port as the sender, **filter** (drop) the frame - the destination is on the same segment.
- If D is unknown, or is broadcast/multicast, **flood** the frame to all ports except the incoming one.

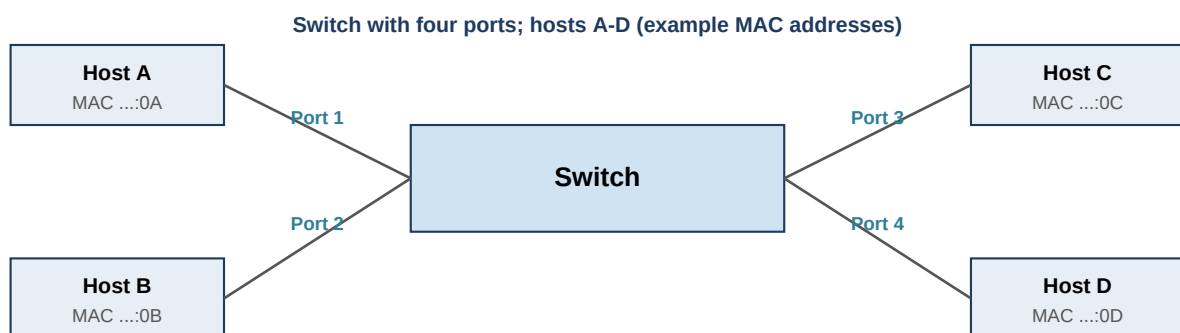


Figure 3: Four-host switched LAN used in the demonstration

Demonstration: building the forwarding table

Step	Event	Switch action	Forwarding table afterwards
0	Switch powered on	-	(empty)
1	A sends frame to B	Learn A on port 1. B unknown, so flood to ports 2, 3, 4	A - 1

Step	Event	Switch action	Forwarding table afterwards
2	B replies to A	Learn B on port 2. A is known on port 1, so forward to port 1 only	A - 1, B - 2
3	C sends frame to A	Learn C on port 3. A known, forward to port 1 only (B and D do not see it)	A - 1, B - 2, C - 3
4	D sends frame to C	Learn D on port 4. C known, forward to port 3 only	A - 1, B - 2, C - 3, D - 4
5	A sends to B	Both known: forward to port 2 only	unchanged (timers refreshed)
6	A broadcasts (ARP)	Destination FF:FF:FF:FF:FF:FF: flood to 2, 3, 4	unchanged

After a few exchanges the table is complete and the switch forwards each unicast frame only where needed. Each port is its own collision domain, so several conversations (A-B and C-D) can proceed **simultaneously**. Entries that are not refreshed within the ageing time are deleted, which also lets the table adapt when a host is moved to another port. Loops among switches are prevented with the Spanning Tree Protocol; otherwise broadcast frames could circulate forever.

Q5. Hub, switch and bridge compared

Feature	Hub (repeater)	Bridge	Switch
OSI layer	Physical (layer 1)	Data-link (layer 2)	Data-link (layer 2); multilayer models also route
Forwarding	Regenerates and repeats the signal on all other ports; no address awareness	Reads MAC addresses; learns and filters between a few LAN segments (software based)	Reads MAC addresses; learns and forwards per port, usually in ASIC hardware
Collision domains	One for the whole hub (all ports share it)	One per attached segment	One per port (with full duplex, no collisions at all)
Broadcast domains	One	One (unless VLAN-aware)	One per VLAN
Ports	Many	Typically 2-4	Many (8 to 48 or more)
Bandwidth	Shared among all hosts, half duplex	Shared within each segment	Dedicated per port, full duplex
Speed / buffering	Same speed on all ports; no buffering	Can connect segments of different types, buffers frames	Store-and-forward or cut-through, mixed speeds, VLAN, QoS, port mirroring
Practical use today	Obsolete; found only in old labs	Joining two LAN segments or media (e.g. wired to wireless); software bridges in virtualisation	Standard LAN building block in offices, labs and data centres; access points include a bridge function

In short: a hub extends a collision domain, a bridge divides a LAN into collision domains using MAC learning, and a switch is a fast multiport bridge that gives each host its own collision domain. Note that many devices are switches without being sold as such: a dual-speed 10/100 Mbps hub is generally a two-port switch (Stevens, ch. 29).

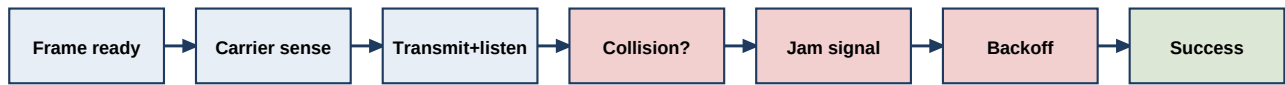
Q6. CSMA/CD and why it matters less today

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the access method of classic half-duplex Ethernet on a shared bus or hub. A station:

- **Senses** the medium; if idle, begins to transmit (if busy, it waits until idle, then sends - 1-persistent).
- **Listens while transmitting** (collision detection). If the signal on the wire differs from what it is sending, a collision has occurred.

- On collision, sends a short **jam signal** so that all stations notice, then stops.
- Waits a random **backoff** of $K \times \text{slot time}$, K chosen from 0 to $2^n - 1$ for the n -th attempt (**binary exponential backoff**, n capped at 10), and retries up to 16 times before reporting failure.

CSMA/CD flow (Ethernet half-duplex)



If idle: transmit. If busy: keep sensing (1-persistent). After a collision: K chosen from $0..2^n - 1$ (n = attempts, max 10).

After 16 failed attempts the NIC gives up and reports an error to the upper layer.

Figure 4: CSMA/CD operation

Why a minimum frame size exists. A sender must still be transmitting when the collision signal returns, so the frame transmission time must be at least twice the maximum one-way propagation time: $T_{fr} \geq 2 T_p$. For 10 Mbps Ethernet this gives a 51.2 microsecond slot time, i.e. 512 bits = 64 bytes, which is why shorter frames are padded.

Why its importance is reduced in modern Ethernet

- **Switches replaced hubs.** Each host connects by its own point-to-point link to a switch port, so there is no shared medium to contend for.
- **Full duplex.** Twisted-pair/fibre links have separate transmit and receive paths. A host can send and receive simultaneously, so collisions cannot occur and CSMA/CD is switched off during auto-negotiation.
- **Gigabit and faster.** 1000BASE-T and above (and 10G, 25G, 100G) are specified only for full-duplex operation in practice; the 64-byte slot-time rule would constrain cable length at such speeds.
- **Switch buffering** handles temporary congestion (frames queue or are dropped) instead of colliding on the wire.
- CSMA/CD remains relevant conceptually, for legacy 10BASE2/10BASE-T hub networks, and as the foundation of access-method study; the same 802.3 frame format is kept for compatibility.

Q7. CSMA/CA in IEEE 802.11 and comparison with CSMA/CD

Wireless stations cannot reliably detect collisions: a radio cannot transmit and listen on the same channel at once, the received signal is far weaker than the transmitted one, and because of the **hidden-station problem** two stations that cannot hear each other may both reach the access point. IEEE 802.11 therefore uses **CSMA/CA** (Collision Avoidance) in its Distributed Coordination Function (DCF) and tries to *avoid* collisions instead of detecting them.

Operation

- **Carrier sense:** the station checks that the channel is idle - physically (energy detection) and virtually (the NAV timer).
- **Inter-frame spacing:** it waits one DIFS; if the channel stays idle it also counts down a random **backoff** taken from the contention window. The counter freezes whenever the channel becomes busy.
- **Transmit** the frame. The receiver, after a short SIFS (shorter than DIFS, giving ACKs priority), returns an **ACK**. No ACK means a collision or loss: the contention window is doubled and the frame retried.
- **Optional RTS/CTS:** for long frames the sender first sends RTS; the receiver replies CTS. Both carry a duration, and every station that hears either sets its **NAV** (network allocation vector) and stays silent. This reserves the channel and solves the hidden-station problem.

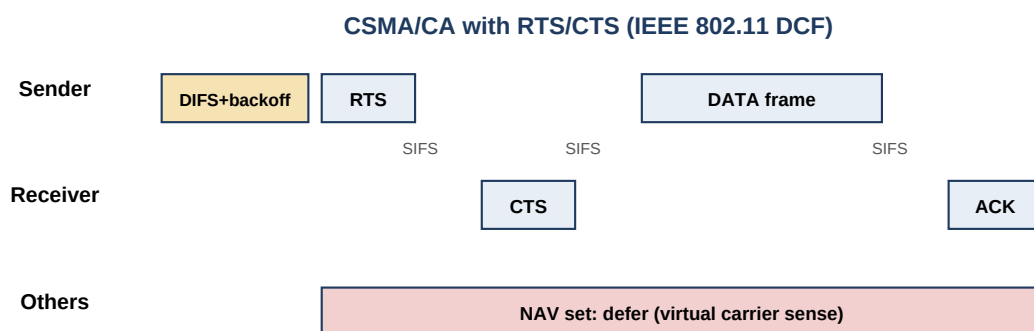


Figure 5: CSMA/CA with RTS/CTS and NAV

Aspect	CSMA/CD (Ethernet)	CSMA/CA (802.11)
Objective	Detect a collision quickly and recover (minimise wasted time)	Avoid collisions before they occur because they cannot be detected
Collision detection	Yes - listen while sending	No - radio is half duplex, hidden stations
Acknowledgement	None at the MAC layer (upper layers recover)	Mandatory ACK per unicast frame
Waiting	Backoff only after a collision	Random backoff <i>before</i> every transmission, after DIFS
Reservation	None	Optional RTS/CTS plus NAV virtual carrier sense
Medium	Wired shared bus/hub (now mostly full duplex)	Shared radio channel
Overhead	Low	Higher (IFS, backoff, ACK, RTS/CTS)

Q8. Wireless LAN design for a college laboratory

Assumptions: one laboratory of roughly 12 m x 9 m with 30 workstations/student laptops, a ceiling at about 3 m, brick walls, and a wired Ethernet backbone with Internet access. A design with two access points gives capacity for around 15 clients each and overlapping coverage for roaming.

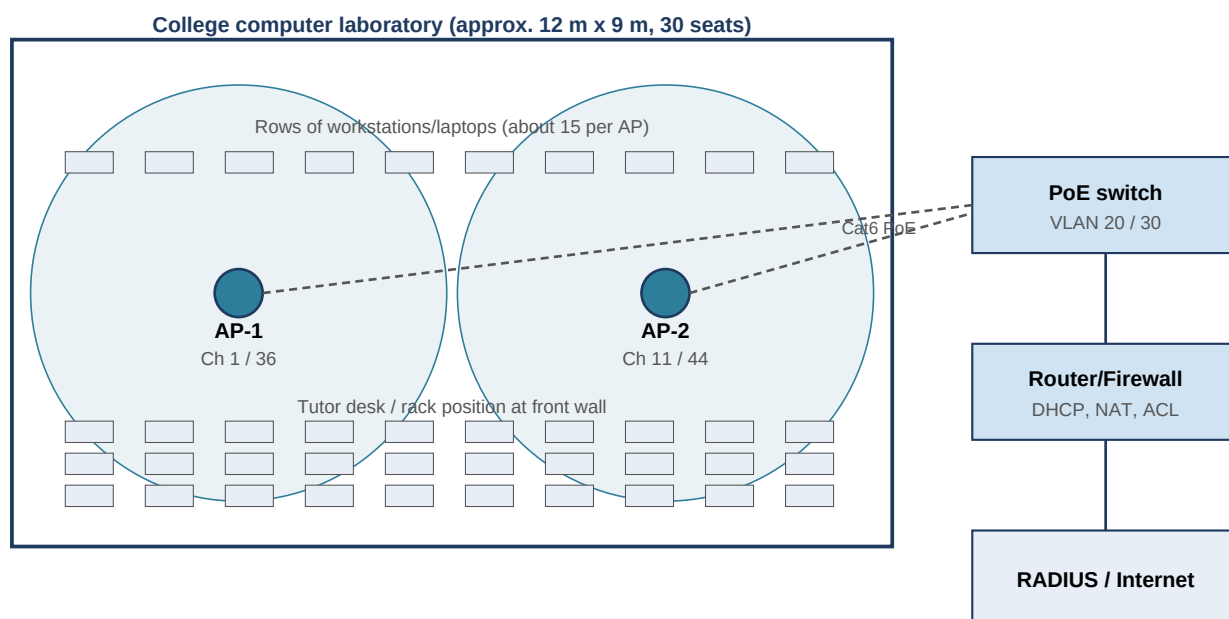


Figure 6: Laboratory WLAN layout (coverage shown by shaded circles)

Item	Design decision	Reason
Access point placement	Two dual-band (2.4/5 GHz) Wi-Fi 6 APs, ceiling-mounted at about 2.7-3 m, one over each half of the lab (about 4 m from the side walls and 6 m apart), powered by PoE	Central, elevated, line-of-sight to desks; about 15 clients per AP; 10-15% cell overlap for roaming; away from metal cabinets and microwave ovens
SSID	CSLab-WiFi (students/staff, 802.1X or WPA3 key) and CSLab-Guest (visitors, isolated). Same SSID on both APs, so one ESS	Same SSID lets devices roam between APs without reconnecting; guest network separated from lab resources
Channel plan	2.4 GHz: AP-1 channel 1, AP-2 channel 11 (non-overlapping; 6 as an alternative). 5 GHz: AP-1 ch 36, AP-2 ch 44 (or 149) with 20/40 MHz width. Transmit power lowered to 30-50%	Adjacent APs on non-overlapping channels avoid co-channel and adjacent-channel interference; 5 GHz gives more capacity; narrower channels suit dense use; band steering moves capable clients to 5 GHz
Addressing	Wired management VLAN 10 (APs: 192.168.10.2-3/24). Client VLAN 20: 192.168.20.0/24, gateway 192.168.20.1, DHCP pool .50-.250, lease 4-8 h. Guest VLAN 30: 192.168.30.0/24, internet-only. Static IPs for APs and printers outside the pool	Separate subnets limit broadcasts and permit different firewall policies; DHCP removes manual configuration; the APs bridge Wi-Fi to the VLANs by MAC (layer 2)
Basic security	WPA3-Personal (SAE) or WPA3/WPA2 mixed mode with a long passphrase; preferably WPA2/WPA3-Enterprise (802.1X, RADIUS) with per-user credentials; disable WPS and open or WEP modes; guest isolation and client isolation; change default admin password, admin over HTTPS/SSH only; keep firmware updated; log and monitor rogue APs	Encryption (AES-CCMP/GCMP) protects traffic over the air; per-user authentication supports accountability; MAC filtering and hidden SSID are not security controls
Capacity / QoS	Limit clients per AP, disable the lowest data rates (e.g. 1-5.5 Mbps), enable WMM	Low legacy rates waste airtime and reduce throughput for everybody
Verification	Site survey with a laptop or phone Wi-Fi analyser; check RSSI better than -67 dBm everywhere and SNR above 25 dB	Confirm coverage, interference and roaming before deployment

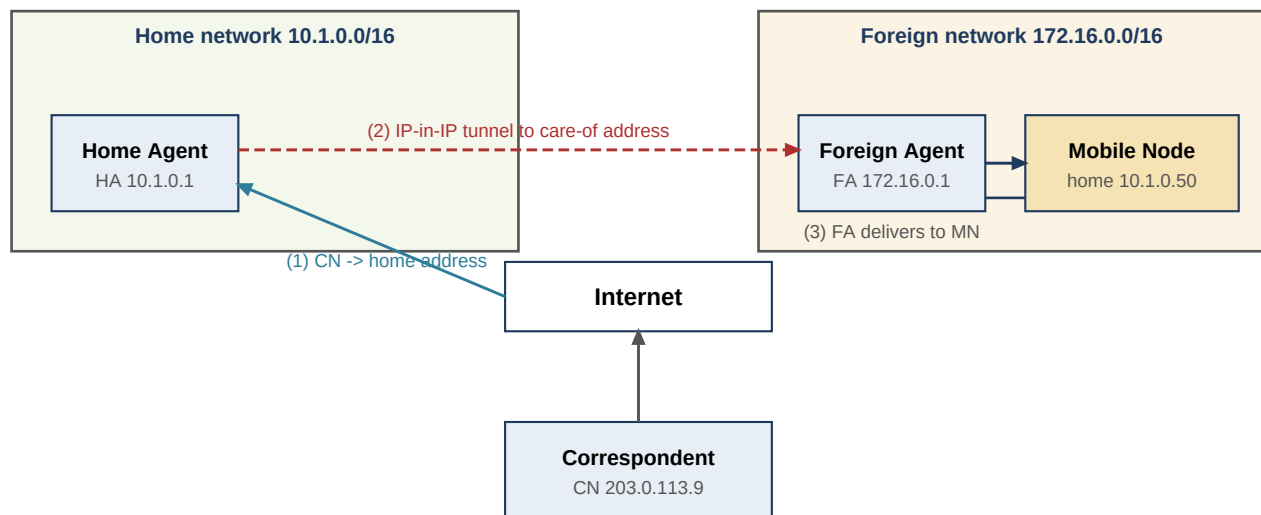
Note: the BSS (one AP and its clients) is identified by the AP's BSSID, which is its MAC address. Two APs sharing one SSID and one wired distribution system form an Extended Service Set (ESS).

Q9. Mobile IP and the need for mobility support

Problem. An IP address has two roles: it identifies a host and indicates the network it belongs to. A router forwards a datagram by the network part of the destination. When a laptop or phone moves to another network, its old IP address is no longer valid there, so packets addressed to it still go to the *old* network. Taking a new IP address would break TCP connections (which are identified by the IP-address and port pairs) and make the host hard to find. **Mobile IP** (RFC 5944 for IPv4) lets a host move between networks while keeping its permanent address, with no change to the correspondent host or to routers.

Components and addressing

- **Mobile node (MN):** the moving host, with a fixed **home address**.
- **Home agent (HA):** a router on the home network that intercepts packets for the MN and tunnels them to it.
- **Foreign agent (FA):** a router on the visited network that provides a **care-of address** and delivers tunnelled packets to the MN (alternatively the MN obtains a co-located care-of address from DHCP).
- **Correspondent node (CN):** the host that communicates with the MN; it keeps using the home address.



Reverse path: MN → FA → CN directly (triangle routing); reverse tunnelling optional. Registration: MN → FA → HA.

Figure 7: Mobile IP operation

Three phases

- **Agent discovery:** HAs and FAs advertise themselves (ICMP router advertisements with a mobility extension); the MN also may solicit. It learns whether it is at home or away and obtains a care-of address.
- **Registration:** the MN (via the FA) registers its care-of address with the HA. The HA builds a binding (home address - care-of address) with a lifetime and authenticates the request.
- **Data transfer:** the CN sends to the home address; the HA intercepts the datagram (proxy ARP), encapsulates it in an IP-in-IP tunnel to the care-of address, and the FA decapsulates it and delivers it to the MN. The MN replies directly to the CN, using its home address as source.

Inefficiency: because inbound traffic goes through the home agent while outbound goes direct, the route can be long (**triangle routing**); the usual remedies are route optimisation and, in IPv6, binding updates.

Why it is needed: it supports always-on connectivity for users moving between campus subnets, Wi-Fi hotspots and cellular networks, without dropping sessions or requiring them to change addresses. Wi-Fi roaming between APs on the *same* subnet is handled at layer 2 and does not need Mobile IP; Mobile IP is for moves across IP networks.

Q10. Packet-level comparison: wired Ethernet vs wireless LAN frame

Method. Capture the wired frame as in Q3. For the wireless frame, use Wireshark on a Wi-Fi interface in **monitor mode** (Linux: `sudo airmon-ng start wlan0` or `iw dev wlan0 set type monitor`; macOS: Wireless Diagnostics or Wireshark's monitor option; Windows normally needs a supported adapter). Use the filter `wlan.fc.type == 2` for data frames, `wlan.fc.type_subtype == 0x08` for beacons and `wlan.fc.type_subtype == 0x1d` for ACKs. Without monitor mode, a Wi-Fi capture shows frames that look like Ethernet, because the driver converts them.

Characteristic	Wired Ethernet II frame	IEEE 802.11 data frame
Wireshark layer	Ethernet II	IEEE 802.11 (plus radiotap header in monitor mode)
Address fields	2 addresses: destination, source (6 B each)	3 addresses (4 in WDS): Addr1 = receiver, Addr2 = transmitter, Addr3 = BSSID or destination/source depending on ToDS/FromDS
Header length	14 B	24 B (30 B with Addr4); 26/32 B with QoS control

Characteristic	Wired Ethernet II frame	IEEE 802.11 data frame
Control information	None (type only)	2-byte Frame Control: version, type (management, control, data), subtype, ToDS/FromDS, Retry, Protected...
Duration/NAV	Not present	2-byte Duration/ID used for the virtual carrier sense
Sequence	None	Sequence Control (fragment and sequence numbers), Retry bit for duplicates
Type of upper protocol	Type/Length field (e.g. 0x0800)	In an LLC/SNAP header inside the body (EtherType again)
Payload size	46-1500 B (jumbo up to ~9000 B)	0-2312 B (typically 1500 B MSDU)
Error check	4-byte FCS, bad frames dropped	4-byte FCS, receiver ACKs good unicast frames; sender retries
Reliability at MAC	None	ACK and retransmission; optional RTS/CTS; fragmentation
Frame kinds	Data only	Management (beacon, probe, auth, assoc), control (RTS, CTS, ACK) and data
Security	None at the link layer (MACsec optional)	Protected bit; payload encrypted with CCMP/GCMP (WPA2/WPA3)
Medium access	Point-to-point full duplex (CSMA/CD only on legacy half duplex)	CSMA/CA, half duplex, shared radio channel

Documented example of the address mapping. For a laptop (STA, MAC 11:11:11:11:11:11) sending to a server beyond the gateway (MAC 33:33:33:33:33:33) through an AP (BSSID 22:22:22:22:22:22):

Hop	Frame type	Address fields
STA to AP (ToDS = 1, FromDS = 0)	802.11 data	Addr1 = 22:...:22 (BSSID/receiver), Addr2 = 11:...:11 (source), Addr3 = 33:...:33 (final destination)
AP to wired network	Ethernet II	Dest = 33:...:33, Src = 11:...:11, Type = 0x0800
Reply: AP to STA (ToDS = 0, FromDS = 1)	802.11 data	Addr1 = 11:...:11 (receiver), Addr2 = 22:...:22 (BSSID/transmitter), Addr3 = 33:...:33 (original source)

The access point converts between the two formats - this is why it is a *bridge* between a wireless and a wired environment, not a switch. Only the link-layer header and trailer change; the IP packet is unchanged. The two screenshots should be saved as screenshots/q10_ethernet_frame.png and screenshots/q10_wlan_frame.png.

Figure 8a: Ethernet frame details - insert screenshot	Figure 8b: 802.11 frame details - insert screenshot
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IEEE 802.11 data frame - field sizes in bytes

Frame Ctrl	Duration	Addr 1	Addr 2	Addr 3	Seq Ctrl	Addr 4	Frame Body	FCS
2	2	6 (RA)	6 (TA)	6	2	6 (opt.)	0 - 2312	4

Frame Control sub-fields (16 bits): Version(2) Type(2) Subtype(4) ToDS FromDS MoreFrag Retry PwrMgmt MoreData Protected Order

Addr 4 appears only when ToDS = 1 and FromDS = 1 (AP-to-AP wireless distribution system).

Addr meanings depend on ToDS/FromDS: e.g. STA->AP: Addr1 = BSSID, Addr2 = source, Addr3 = destination.

Figure 9: IEEE 802.11 frame format (for comparison with Figure 1)

References

- B. A. Forouzan, *Data Communications and Networking / Computer Networks* (TCP/IP protocol suite), ch. 5 "Data-Link Layer: Wired Networks" and ch. 6 "Wireless Networks and Mobile IP" (course lecture notes: Datalink_Layer_1.pdf, Datalink_layer_2.pdf).
- W. R. Stevens, B. Fenner, A. M. Rudoff, *UNIX Network Programming, Vol. 1*, ch. 29 "Datalink Access" (course case study: DL_Case_Study.pdf).
- IEEE Std 802.3 (Ethernet) and IEEE Std 802.11 (Wireless LAN) - standards.
- C. Perkins (ed.), RFC 5944: *IP Mobility Support for IPv4, Revised*, IETF, 2010.
- D. Plummer, RFC 826: *An Ethernet Address Resolution Protocol*, 1982.
- Wireshark User's Guide and display filter reference, <https://www.wireshark.org/docs/>.